중적분 14

14.5~14.6 원통(구면)좌표계상의 3중적분

2. (1) $\int_0^{2\pi}\int_1^2\int_0^5 e^z\, r\, dz dr\, d\theta$

[Sol] $\int_0^{2\pi}\int_1^2\int_0^5 e^z\, r\, dz dr\, d\theta = \int_0^{2\pi}\int_1^2 r(e^5-1)\, dr\, d\theta$ 「$\because \int_0^5 e^z r\, dz = r\left[e^z\right]_0^5 = r(e^5-1)$ 」

$= \int_0^{2\pi} \frac{3}{2}(e^5-1)\, d\theta$ 「$\because (e^5-1)\int_1^2 r\, dr = (e^5-1)\left[\frac{1}{2}r^2\right]_1^2 = \frac{3}{2}(e^5-1)$ 」

$= 3\pi(e^5-1)$

2. (2) $\int_0^{\pi/2}\int_0^1\int_0^{\sqrt{1-r^2}} r\sin\theta\, dz dr\, d\theta$

[Sol] $\int_0^{\pi/2}\int_0^1\int_0^{\sqrt{1-r^2}} r\sin\theta\, dz dr\, d\theta = \int_0^{\pi/2}\int_0^1 r\sqrt{1-r^2}\sin\theta\, dr\, d\theta$

「$\because u = 1 - r^2 \Rightarrow du = -2r\, dr$

$$\int_0^1 r\sqrt{1-r^2}\sin\theta\, dr = \int_1^0 \sqrt{u}\sin\theta\left(-\frac{1}{2}\right)du$$

$$= \frac{1}{2}\sin\theta\int_0^1 \sqrt{u}\, du$$

$$= \frac{1}{2}\sin\theta\left[\frac{2}{3}u^{\frac{3}{2}}\right]_0^1 = \frac{1}{3}\sin\theta$$ 」

$$= \int_0^{\pi/2}\frac{1}{3}\sin\theta\, d\theta$$

$$= \frac{1}{3}\left[-\cos\theta\right]_0^{\frac{\pi}{2}} = \frac{1}{3}$$

2. (3) $\iiint_D zy\,dV$, 여기서 D는 평면 $z=1$의 위쪽과 원뿔 $z=\sqrt{x^2+y^2}$의 아래쪽에 의해 둘러싸인 입체영역

[Sol] $\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{1}^{\sqrt{x^2+y^2}} zy\,dzdydx$ 「$\because \displaystyle\int_{1}^{\sqrt{x^2+y^2}} zy\,dz=\frac{y}{2}\left[z^2\right]_{1}^{\sqrt{x^2+y^2}}=\frac{y}{2}(x^2+y^2-1)$ 」

$$=\int_0^{2\pi}\int_0^1 \frac{r\sin\theta}{2}\{r^2-1\}r\,drd\theta$$

「$\because \displaystyle\int_0^1 \frac{r\sin\theta}{2}\{r^2-1\}r\,dr=\frac{\sin\theta}{2}\int_0^1 r^4-r^2\,dr$

$$=\frac{\sin\theta}{2}\left[\frac{1}{5}r^5-\frac{1}{3}r^3\right]_0^1$$

$$=-\frac{1}{15}\sin\theta$$ 」

$$=\int_0^{2\pi}-\frac{1}{15}\sin\theta\,d\theta$$

$$=-\frac{1}{15}\left[-\cos\theta\right]_0^{2\pi}$$

$$=\frac{1}{15}\left[\cos\theta\right]_0^{2\pi}=0$$

3. (1) $\iiint_D (x^2+y^2)\,dV$, 여기서 D는 원통 $x^2+y^2=1$과 평면 $z=0$, $z=4$에 의해 둘러싸인 입체영역

[Sol] $\iiint_D (x^2+y^2)\,dV = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_0^4 x^2+y^2\,dz\,dy\,dx$

$$= \int_0^{2\pi}\int_0^1\int_0^4 (r^2)\,r\,dz\,dr\,d\theta \qquad 「\because \int_0^4 r^3\,dz = \left[r^3 z\right]_0^4 = 4r^3 」$$

$$= \int_0^{2\pi}\int_0^1 4r^3\,dr\,d\theta \qquad 「\because \int_0^1 4r^3\,dr = \left[r^4\right]_0^1 = 1 」$$

$$= \int_0^{2\pi} 1\,d\theta = 2\pi$$

3. (1) $\int_0^{2\pi}\int_0^{\pi/4}\int_0^1 \rho^2 \sin\phi \, d\rho d\phi d\theta$

[Sol] $\int_0^{2\pi}\int_0^{\pi/4}\int_0^1 \rho^2 \sin\phi \, d\rho d\phi d\theta$

「$\because \sin\phi \int_0^1 \rho^2 \, d\rho = \sin\phi \left[\frac{1}{3}\rho^3\right]_0^1 = \frac{1}{3}\sin\phi$ 」

$= \int_0^{2\pi}\int_0^{\pi/4} \frac{1}{3}\sin\phi \, d\phi d\theta$

「$\because \frac{1}{3}\int_0^{\pi/4} \sin\phi \, d\phi = \frac{1}{3}[-\cos\phi]_0^{\pi/4} = -\frac{1}{3}\left[\cos\frac{\pi}{4} - \cos 0\right] = -\frac{1}{3}\left(\frac{\sqrt{2}-2}{2}\right)$ 」

$= \int_0^{2\pi} -\frac{1}{3}\left(\frac{\sqrt{2}-2}{2}\right) d\theta$

$= -\frac{1}{3}\left(\frac{\sqrt{2}-2}{2}\right)[\theta]_0^{2\pi} = \frac{\pi}{3}(2-\sqrt{2})$

3. (2) $\int_0^{2\pi}\int_0^{\pi/2}\int_1^3 \rho^3\cos\phi\sin\phi\ d\rho d\phi d\theta$

[Sol] $\int_0^{2\pi}\int_0^{\pi/2}\int_1^3 \rho^3\cos\phi\sin\phi\ d\rho d\phi d\theta$

「$\because\ \cos\phi\sin\phi\int_1^3 \rho^3\ d\rho = \cos\phi\sin\phi\left[\frac{1}{4}\rho^4\right]_1^3 = \cos\phi\sin\phi\left[\frac{81-1}{4}\right] = 20\cos\phi\sin\phi$ 」

$= \int_0^{2\pi}\int_0^{\pi/2} 20\cos\phi\sin\phi\ d\phi d\theta$

「$\because\ u = \sin\phi \Rightarrow du = \cos\phi d\phi$

$20\int_0^{\pi/2}\cos\phi\sin\phi\ d\phi = 20\int_0^1 u\ du = 20\left[\frac{1}{2}u^2\right]_0^1 = 10$ 」

$= \int_0^{2\pi} 10\ d\theta = 10[\theta]_0^{2\pi} = 2\pi$

3. (3) $\int_0^{\pi}\int_0^{\pi/2}\int_0^{\sin\phi}\rho^2\sin\phi\, d\rho d\phi d\theta$

[Sol] $\int_0^{\pi}\int_0^{\pi/2}\int_0^{\sin\phi}\rho^2\sin\phi\, d\rho d\phi d\theta$

「$\because \sin\phi\int_0^{\sin\phi}\rho^2\, d\rho=\sin\phi\left[\frac{1}{3}\rho^3\right]_0^{\sin\phi}=\frac{1}{3}\sin^4\phi$ 」

$=\int_0^{\pi}\int_0^{\pi/2}\frac{1}{3}\sin^4\phi\, d\phi d\theta$

「$\because \frac{1}{3}\int_0^{\pi/2}\sin^4\phi\, d\phi=\frac{\pi}{16}$ P627 (15)

$\int\sin^4\phi\, d\phi=\frac{3}{8}\phi-\frac{1}{4}\sin(2\phi)+\frac{1}{32}\sin(4\phi)$ or $-\frac{1}{4}\sin^3\phi\cos\phi-\frac{3}{8}\sin\phi\cos\phi+\frac{3}{8}\phi$ 」

$=\int_0^{\pi}\frac{\pi}{16}\, d\theta=\frac{\pi}{16}[\theta]_0^{\pi}=\frac{\pi^2}{16}$

4. (1) $\iiint_D (x^2+y^2)\,dV$, D는 $x^2+y^2+z^2 \le 1$인 구이다.

[Sol] $\iiint_D (x^2+y^2)\,dV = \int_0^{2\pi}\int_0^{\pi}\int_0^1 \{(\rho\sin\phi\cos\theta)^2+(\rho\sin\phi\sin\theta)^2\}\rho^2\sin\phi\,d\rho\,d\phi\,d\theta$

「$\because \int_0^1 \rho^4\sin^3\phi\,d\rho = \sin^3\phi\int_0^1\rho^4\,d\rho = \sin^3\phi\left[\frac{1}{5}\rho^5\right]_0^1 = \frac{1}{5}\sin^3\phi$ 」

$= \int_0^{2\pi}\int_0^{\pi}\frac{1}{5}\sin^3\phi\,d\phi\,d\theta$

「$\because \int_0^{\pi}\frac{1}{5}\sin^3\phi\,d\phi = \frac{1}{5}\int_0^{\pi}(1-\cos^2\phi)\sin\phi\,d\phi$

$u=\cos\phi \Rightarrow du = -\sin\phi\,d\phi$

$= \frac{1}{5}\int_1^{-1} 1-u^2\left(\frac{1}{-1}\right)du = \frac{1}{5}\int_{-1}^{1} 1-u^2\,du$

$= \frac{2}{5}\int_0^1 1-u^2\,du = \frac{2}{5}\left[u-\frac{1}{3}u^3\right]_0^1 = \frac{4}{15}$ 」

$= \int_0^{2\pi}\frac{4}{15}\,d\theta = \frac{4}{15}[\theta]_0^{2\pi} = \frac{8}{15}\pi$